

6634	Open	POWERLINE EMBAYMENT: A) 1975 Summary report of exploration. B) 1977 Geophysical logging of PL32.
6635	"	POWERLINE EMBAYMENT: Drill logs, PL SERIES, i.e. PL1 to PL47 inclusive.
6636	"	BOOKALOO-LAKE DUTTON: Geophysical logging of drillholes, LD series BK series.
6637	"	BOOKALOO-LAKE DUTTON-LAKE MACFARLANE: Drill logs: Bookaloo BK1-BK22 Lake Dutton LD8-LD38 Lake MacFarlane MF1-MF13.
6638	"	LY SERIES DRILLING: A) LY2 (Cattlegrid) 1974 Geophysical logs. B) LY4-LY11 and 83/108 (East of Cattlegrid), 1983 Geological logs.
6639	"	BIOGEOCHEMISTRY: A) CATTLEGRID & MG14 AREAS (Pacminex 1978). B) PERNATTY LAGOON (Austminex, 1966).
6640	"	GROUNDWATER GEOCHEMISTRY - Stuart Shelf: Pacminex (1978) - Report PMR90/78.
6663	"	MT. WHYALLA BARITE DEPOSIT: Pacminex (1974) report - PMR 57/74.
6664	"	GEOLOGICAL MAPPING: Report: Elizabeth Creek area (Chisholm, 1974) Plans: Pernatty Lagoon Mt. Gunson Mines area Elizabeth Area Canegrass Dam area Wirrappa area.

6666 Open

DRILL LOGS - ELIZABETH CREEK & CANEGRASS AREAS:

- EC23-EC54 (Elizabeth Creek)
- TOC1-TOC4
- SP1-SP19 (Snakepit)
- 3-14-19P to 3-14-41P (Canegrass Dam area)
- 3-14-1P to 3-14-46P (Canegrass field logs).

6667

DRILL LOGS-PANDURRA AREA (EL405):

- Vol. 1. EX1-EX157
2. EX158-EX163 & YD1-YD15
3. EX164-EX184.
4. Area 1:100 000 Drillholes
- | | | |
|-------------|------|---------|
| | 6332 | J12-J24 |
| Roopena | " | P1-P4 |
| " | " | J1-J11 |
| " | " | I1 |
| Cultana | 6432 | Q1, Q2 |
| Cariewerloo | 6333 | Y1-Y6 |
| " | " | T1-T5 |
| " | " | X1-X3 |

6668

DRILL LOGS-WATER BORES in the MT. GUNSON area.

6669

DRILLING SUMMARIES:
3-1-X to 3-14-X series.

6670

DRILL LOGS-LAKE TORRENS AREA:
(see also ENV. 3769)
BDH1-BDH4
SLT101-SLT107
YAD1.

6674

- ELECTROMAGNETIC SURVEYS:
- A) Gonyah Lake, 1971 hand-held survey (1 plan only)
 - B) Cattlegrid & MGL4 areas, airborne EM INPUT Test Survey by Geoterrex, 1973.
 - C) Winnie Pinnie area & Yudnapinna area, airborne electromagnetic survey, Barringer INPUT system, 1975. Geoterrex report 83-253 PMR Report 144/75.
 - D) Stuart Shelf (EL186, EL199), airborne EM INPUT survey by Geoterrex, 1975.
 - E) Mount Gunson Pulse Electromagnetic Test, 1977. (Ground Survey) PMR Report 58/77. Cattlegrid anomaly & Anomaly 'J' (margin of Pernatty Lagoon).
 - F) Pandurra & Cariewerloo, airborne EM INPUT Survey by Geoterrex, 1977.

6675

- REFRACTION SURVEYS:
- A) Austral Exploration (1967)
Refraction Test Survey at Main Open Cut;
West Lagoon; East Lagoon.
 - B) Austral Exploration (1973)
Reconnaissance Seismic Refraction Survey -
Mt. Gunson area.

6676

Open

GRAVITY SURVEYS: GRAVITY SURVEYS

- A) Gravity survey - Pandurra area, EL186. Survey by Solo Geophysics, PMR114/77.
- B) Gravity anomaly 'A' - Site for drillhole LHL. Notes (1978).
- C) Magnetic & gravity survey with levelling at Illeroo Grid (Pandurra area, EL534. Report by Solo Geophysics (1979), PMR87/79.
- D) Gravity Survey with barometric levelling at Yudnapinna Reconnaissance Line, EL534. Report by Solo Geophysics (1979), PMR88/79.
- E) Interpretation of Geophysics, Illeroo area, EL534. Report by Langron (1980) PMR10/80.
- F) Gravity survey - Pernatty Lagoon area, EL543. Survey by Solo Geophysics (1980).
- G) Gravity survey - Mt. Gunson area, 'Blue Mag' anomaly 1981.

6677

AEROMAGNETIC SURVEYS:

- A) Report on stage 1 on the interpretation of results of airborne magnetometer survey covering SML139, 152.
- B) Report on an aeromagnetic interpretation, Torrens map sheet, S.A. PMR80/75.
- C) Plan, EL186, April 1977. Magnetic Profiles.
- D) Geoex (1978) Survey.
Plans of: Aeromagnetic total intensity
: Profiles.
- E) Report on airborne magnetic surveys over an area East of Lake Dutton (Pernatty & Bookaloo 1:63 360 sheets). PMR10/78.

6678

RESISTIVITY SURVEYS:

- A) Resistivity Survey at Pandurra near Port Augusta for Pacminex Pty. Ltd. EL186. Report No. 265 by Murdoch Geophysics, 1977. PMR60/77.
- B) A resistivity survey at Mt. Gunson, S.A. for Pacminex. (Survey in June 1976) EL199. Report No. 266 by Murdoch Geophysics, 1977. PMR142/76.
- C) Evaluation of a resistivity survey at Mt. Gunson, June 1976. PMR173/76, EL199.
(Localities: Cattlegrid; Oakden Hills; MG14).
- D) Geophysical Survey at Cattlegrid, July 1973. Survey by CGG; PMR report 133/73.

6679

MISCELLANEOUS GEOPHYSICAL SURVEYS - MT. GUNSON AREA:

- A) Report on a combined Induced Polarisation, Seismic and EM survey in the Mt. Gunson area. MacPhar Geophysics (1973).
- B) Gravity and magnetic surveys on the North Pernatty Grid.
Lines 6,530,000N to 6,540,000N.
Solo Geophysics (1981).
- C) Magnetic surveys of 'Blind Dyke', 1982-1983. Un-compiled plans and notes. Includes data of Solo Geophysics (1982).

6680

GEOPHYSICAL LOGGING OF DRILLHOLES-STUART SHELF:

- A) Geophysical Logging of Drillhole PL32, Gunson area (i.e. Powerline Embayment). PMR138/77.
- B) Geophysical logging of drillholes in the Mt. Gunson and Pandurra areas. PMR59/78.
 - i.e. OK117-OK120
 - OK128-OK129
 - WPB2-WPB4
 - WPB6
 - WOB10-WOB13
 - LW34, LW38, LW40
 - PL34
 - MG-11-3
- PANDURRA AREA:
 - J12, J15, J16, J20, J21.
- C) Geophysical logging of percussion drillholes, June-August 1978. PMR89/78. (EL332,338,339),
 - i.e. EC5
 - PL40
 - LD31
 - BK10-BK12
 - EX158-EX163
- D) Geophysical logging in the lake Windabout, Powerline and Oakden Hills areas, EL339,
 - i.e. LW61-LW75
 - PL42-PL44
 - OK132-OK135
 - PMR Report 36/79.
- F) Gamma logs - assorted drillholes.

6681

INDUCED POLARISATION SURVEYS:

- 1) IP Survey by Geoscience (1965) for Austminex, SML55.
- 2) IP and test resistivity survey by MacPhar Geophysics (1967-1968) for Austminex.
- 3) IP survey by Austral Exploration, 1971-1972, at:
 - Main Open Cut
 - Gunyah Lake
 - East Lagoon
 - West Lagoon
 - Mystery
 - Sweet Nell areas.
- 4) Multi-mode I.P. survey by McPhar Geophysics, 1974 at:
 - Cattlegrid
 - Gunyah LakePMR report 75/74.

6682

INDUCED POLARISATION SURVEY:

by SOLO GEOPHYSICS, 1983.
CATTLEGRID - MAIN OPEN CUT AREAS,
Includes: 'Haul Road' anomaly
'Dyke' anomaly.

6483

OPEN

regional plans and reports. 'MOUNT GUNSON area' - regional plans and reports.

Includes:

- A) Austminex (1966-1968) reports for SML 96, 121 and 139 - Pernatty Lagoon area.
- B) SADME articles on Mt. Gunson - extracted from Mining Reviews.

6714	"	MAIN OPEN CUT - plans
6715	"	CATTLEGRID - cross-sections.
6718	OPEN	HOUSE WORKINGS and GUNYOT workings. Plans only - sections, ore reserves, grade control and hole locations. Austminex Pty Ltd and Mount Gunson Mines Pty. Ltd.
6721	OPEN	CATTLEGRID: Surveying and engineering plans, pit development 1974-1986. Includes 1977 aerial survey.
6722	"	TOWN, CARAVAN PARK, SERVICES & TAILINGS DAM - surveying and engineering plans.
6723	"	CATTLEGRID - grade control for areas C2, C3 and Area IV. Predominantly plans - blasthole layout, top-of-ore, grade control plans and cross-sections.
6724	"	EAST LAGOON OREBODY - plans
6725	"	NORANDA AUSTRALIA Ltd (1967). Plans - Main Open Cut, House Workings, West Lagoon, East Lagoon, Mystery.
6726	OPEN	MAIN OPEN CUT - Cross-sections by Austminex Pty Ltd (1967-1968).
6727	"	WEST LAGOON - plans by Mount Gunson Mines and Austminex Pty Ltd. (Grade control, block plans, face geology, drillholes, cross-sections etc.)
6728	"	CATTLEGRID - Grade control plans for areas B2, B3 and B4 (east-northeast Cattlegrid). Plans of bottom-of-ore, top-of-ore, diamond drill-holes, diamond drilling influence areas as well as pit - and grade control - cross-sections.
6729	"	CATTLEGRID - End-of-period Survey Plans (1980-1982).

TOTAL: - 73 open file

ENVELOPE 2330
EXPLORATION LICENCE 81

(Page 1).

Report:

Quarterly report, period ending 19.10.73
(PMR 105/73)

Plans:

<u>Drawing No.</u>	<u>Project</u>	<u>Title</u>	<u>Date</u>
1551	Birthday - Mt. Moseley	Geology	1.9.73
47	Pornatty & Island Lagoon.	Regional Geology	10.10.72
1565	EL 81	Bedrock Lithology	1.10.73
1552.	Birthday - Mt. Moseley	" " "	1.9.73
1521 A	Sheet 1	Outcrop & bedrock geology	1.11.73
1520 A	Cattlegrid, Sheet 5	Drillhole Location plan	1.9.73
1526 A	" —	Cross Section 6,518,450 mN	1.11.73
" B	"	" " 6,518,700 mN	"
" C	"	" " 6,518,900 mN	"
" D	"	" " 6,519,100 mN	"

Report:

Results of Exploration activities in the Oakden Hills -
Lake MacFarlane area.

Maiden, K.J.

Sept. 1973

(PMR 97/73)

DRILL LOGS αxxx/xxx series.
" " MG series. (MG 14 area)
" " OK series i.e. OK 1 → 12.
(Oakden Hills area)
" " Lake MacFarlane area.
MF 1 → 6.

Plans:

<u>Drawing No.</u>	<u>Project</u>	<u>Title</u>	<u>Date</u>
1554	Birthday - Mt. Moseley	Geology & copper geochem. plan	1.9.73

ENV. 2330 (cont.)

(Page 2)

<u>Drawing No.</u>	<u>Project</u>	<u>Title</u>	<u>Date</u>
1562	Oakden Hills	Aeromag. Interpretation -	
1553	Birthday - Mt. Mosely	Structural and palaeogeographic oakden hills interpretation.	1.9.73
1567	Cattlegrid	Geophysical layout & principal results	1.10.73

Volume III

Quarterly report for EL 81, period ending 19.1.74.
(PMR 49/74)

includes:

orientation geophysical tests (August, Sept. 1973).
Langron, W.J. (PMR 107/73)

Plans:

<u>Drawing No.</u>	<u>Project</u>	<u>Title</u>	<u>Date</u>
1568	Mt. Gunson	Locality plan & input flight lines.	1.10.73
1564	AIRSTRIP area.	Magnetic profiles (total field) airstrip area	"
1561	Cattlegrid area.	Ground magnetic profiles	"
1560	"	S.P. Profiles	"
1558	Airstrip area.	" "	"
1559.	MG 14	Magnetic profiles (total field)	"
1557	"	S.P. Profiles	"
1569	Mt. Gunson	Input Contours, channel 4	"

Volume IV :

Report:

Report on a combined induced polarisation
survey in the Mt. Gunson area, S.A. for
Palmirex Pty Ltd.

PMR 114/73

by McPhar Geophysics.

Reports: Quarterly reports, periods ending 19.4.74 (PMR 53/74)
 " 20.7.74 (PMR 91/74)

Includes drill logs for:

PL 1

OK 27 - 48

L72, L73

Report: L73 drill core petrography, Cattlegrid. (PMR 62/74)

" Geoterrex input test survey at Mount Gunson
 for Peemirex Pty Ltd (Part II of III)

J.E. Hansen.

(PMR 100/73)

— with Geoterrex plans —

Plans:

<u>Drawing No.</u>	<u>Project</u>	<u>Title</u>	<u>Date.</u>
2030	Oakden Hills	Geochemical Survey (Cu)	1.7.74
1525 E	Pernatty Lagoon	Palaeogeographic Interpretation	1.4.74
2031	Powerline	Input anomaly C	

Volume VI:

Report: Quarterly report for period ending 19.10.74.
 (PMR 135/74)

DIAMOND DRILL LOGS MG 15D — MG 41

POWERLINE PL 1-6.

Volume VII

Drill logs - MG 24, MG 53.

" " - Oakden Hills OK 13 - 27.

OK 49.

" " Power Line series.

" " — north of Cattlegrid. 704-Xseries
 ie. 704-1 to 704-29

" " — Oakden Geochem. grid.
 OK 3-1 to OK 3-56.

ENV. 2330 (cont)

Page 4.

Volume VIII :

Report: Quarterly report to 19.1.75 PMR 13/75

Includes plans:

<u>Drawing No.</u>	<u>Project</u>	<u>Title</u>	<u>Date</u>
300 B	Mount Gunson	Location & results of I.P. & Geochem. traverses DIAMOND DRILL LOGS — Lake MacFarlane area i.e. MF 7 — MF 13	11.4.72

Report: LY3 DRILL CORE PETROGRAPHY,
CATTLEGRID, PART II. (PMR 115/74)

P.J. CURTIS

FURTHER STUDY OF ALTERED ACID VOLCANIC
FROM BASE (~ 246m) OF CATTLEGRID LY3
DIAMOND DRILL HOLE, MOUNT GUNSON.

P.J. CURTIS

PMR 143/74

QUARTERLY REPORT for PERIOD ENDING 20.4.75.

PMR 45/75

Volume IX :

Report: Final Quarterly report, period ending 20.7.75.

PMR 132/75

Includes: ① Diamond drill logs — MG14 area
i.e. MG 54 to MG 68

② " " " — GULLY area.
i.e. GY 1 to GY 16.

③ Interpretation report, airborne EM survey,
Barringer 'INPUT' system of the
Lake Windabout, Oakden Hills, Lake Sutton
areas.

by J. E. Hansen, Geotrex Ltd. PMR 93/75

ENV 2330 (cont.)

(page 5)

- ④ Report on an aeromagnetic interpretation
- ~~Torrens~~ map sheet
by J.L. Irvine

PMR 80/75

MOUNT GUNSON MINES PTY LTD

QUARTERLY REPORT

ON EXPLORATION LICENCE NO. 81,

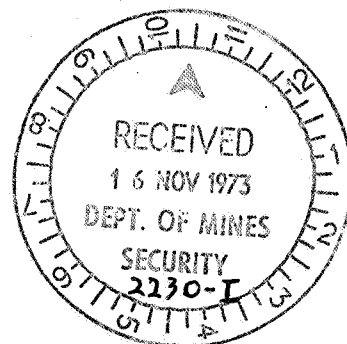
MT. GUNSON

FOR THE PERIOD ENDING

19TH OCTOBER, 1973

MT. GUNSON
October, 1973.

J.M. HEAPE



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1.3	Geochemistry	1
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- I RESULTS OF EXPLORATION ACTIVITIES IN THE
OAKDEN HILLS-LAKE MACFARLANE AREA
- II GEOPHYSICAL ACTIVITIES IN E.L. 81
- III DRILL LOGS

LIST OF ILLUSTRATIONSPLANSPMX DWG NO.

- 1565 BEDROCK LITHOLOGY OF E.L. 81 (1:125,000)
- 1521A OUTCROP AND BEDROCK GEOLOGY, MT. GUNSON
AREA (1:10,000)
- 1520A DRILL HOLE LOCATION PLAN, CATTLE GRID
(1:2,500)
- 1526A/D GEOLOGICAL CROSS SECTIONS, CATTLE GRID

1. SUMMARY

This report covers work carried out on E.L. 29 from 8th June to 19th July, 1973, and subsequently on E.L. 81, from 20th July to 19th October, 1973. E.L. 81 is an amalgamation incorporating E.L. 29.

During this period exploration activity has consisted of :-

1.1 Drilling

42 holes were drilled totalling 2,190 metres including 171 metres of diamond drilling.

1.2 Geological Mapping

This was carried out in the Mt. Moseley and Lake Windabout areas (refer Appendix I).

1.3 Geochemistry

Further evaluation of samples from drill holes over the Cattle Grid deposit delineated an anomaly at shallow depths over the deposit. On the basis of this orientation work on a grid pattern of shallow drill samples may show up targets of possible mineralisation at depth. Grid lines are now being laid out in preparation for such a shallow geochemical drilling programme.

1.4 Geophysical Surveys

Orientation surveys over the undisturbed Cattle Grid copper deposit have been undertaken (refer Appendix II).

1.5 Mineral Claims

Mineral Claims have been pegged over the Cattle Grid copper deposit in preparation for mining. Data on drilling and shaft sinking within these Mineral Claims is still being processed and analysed and is not referred to in this report.

2. TENEMENTS AND REPORTING

2.1 Exploration Licence Title Position

On 20th July, 1973, E.L. 29 was amalgamated with adjoining licence areas applied for by Mount Gunson Mines, to become part of E.L. 81.

This report covers work carried out on E.L. 29 from 8th June to 19th July, 1973, and subsequent work carried out on E.L. 81, for the first quarterly period ending 19th October, 1973.

2.2 Mineral Claims Over Cattle Grid Deposit

Mineral Claim Nos. 251 to 279 were granted to Mount Gunson Mines Pty Ltd on 7th August, 1973. These claims cover the recently outlined Cattle Grid copper deposit. It is the intention of Mount Gunson Mines to continue feasibility studies with a view to mining the Cattle Grid orebody if sales contracts are negotiated and approved by the South Australian and Australian Governments.

3. EXPLORATION ACTIVITY

3.1 Geological Mapping 1:25,000

3.1.1 Mt. Moseley-Oakden Hills Area

A geological map with an accompanying report has been prepared for this area. Drilling was necessary to gain subsurface information below the areas of sand cover.

Geochemical surface sampling of rock outcrops showed interesting copper values which will be further investigated to try and define the source of the anomalous values.

3.1.2 Lake Windabout Area

Field mapping commenced in the area west of Lake Windabout, and will be continued eastwards to Bottle Hill on the eastern boundary of E.L. 81. Airphoto interpretation has already been completed over this area.

3.1.3 Cattle Grid

Sections across the Cattle Grid orebody have been prepared and are enclosed (Appendix III).

Diamond core from the area has been independently relogged by Pacminex geologist, P.M. Macnamara, who also mapped both shafts sunk into the deposit.

3.1.4 Lake MacFarlane Area

Some attention has been paid to large areas of gypsiferous strata within the E.L. 81 to obtain a preliminary view of its potential.

3.2 Geochemistry

Further study has been made of anomalous values in superficial strata over the Cattle Grid deposit.

It would offer from this study that a geochemical drilling programme will delineate deep mineralisation if conditions similar to those in the Cattle Grid deposit are repeated. It is hoped to commence a grid pattern drilling programme (to 15 m depth) as soon as an effective drilling rig can be obtained.

As a preliminary to this work, grid lines initially 1,000 metres apart are being laid down, north, south and west of the previously surveyed grid. Approximately 45 km of lines have so far been pegged.

3.3 Geophysics

Orientation geophysical surveys were carried out during the period 18th to 31st July, 1973, and again from the 31st August to 14th September, 1973. These surveys were carried out over the Cattle Grid deposit to test the applicability of various geophysical techniques on an established orebody before disturbance by open cast mining.

3.3.1 Geophysical Methods

- (i) Ground Magnetics.
- (ii) Self Potential.
- (iii) Induced Polarisation including pulse pole-dipole, dipole-dipole, a saturation test down two drill holes and Mise-a-la-masse down one drill hole.
- (iv) INPUT (airborne E.M.).
- (v) Hammer Seismic.
- (vi) 2 Coil Electro-Magnetics.

The results of surveys are still to be interpreted in detail and final reports await this work. A separate summary (by the Chief Geophysicist, Pacminex) is appended (Appendix II).

3.3.2 Preliminary Results

- (i) Ground Magnetics : some small anomaly effects may be due to mineralisation but a general low is not apparent.

- (ii) S.P. : optimum ground conditions prevailed during the S.P. work. Some small, distinct negative anomalies were produced. However there appears to be no consistent relationship to the mineralisation.
- (iii) I.P. : results have not yet been processed but some good responses do occur over portions of the orebody.
- (iv) INPUT : several recognisable anomalies occur over the Cattle Grid, however results are still to be fully assessed before any conclusions can be drawn.
- (v) Seismic : this method is ideally suited to detect the layered structure within the sediments but the energy source is operating at or beyond its reliable limit.
- (vi) E.M. : because of low surface resistivities the data was very noisy. All anomalies detected were attributed to surface conductors. This method is therefore not considered suitable for reconnaissance.

3.4 Drilling

The drilling carried out during this period had three primary aims :-

- (i) wide spaced drilling for regional stratigraphic information;
- (ii) close spaced drilling looking for repetitions of the Cattle Grid deposit;
- (iii) close spaced in-fill and delineation drilling over the Cattle Grid deposit.

That relating to (i) and (ii) were primarily rotary drill holes with limited coring for rock identification, whereas that relating to (iii) consisted of rotary drilling for diamond drill precollaring and only the ore zone was diamond drilled.

3.4.1 Summary of Drilling Data

Rotary drilling of (i) and (ii) above.

Number of Holes Drilled = 28

Metres Drilled = 1,637 m

Diamond drilling of (i) and (ii) above.

Number of Holes Drilled = 1

Metres Drilled = 4.90 m

Rotary Drilling (precollaring Cattle Grid).

Number of Holes Drilled = 13

Metres Drilled = 388 m

Diamond Drilling (Cattle Grid).

Number of Holes Drilled = 13

Metres Drilled = 166 m

Log sheets for these drill holes are attached to this report (Appendix III).

3.4.2 Surveying of Drill Holes

Sites of all diamond and rotary drill holes adjacent to the Cattle Grid area have been surveyed by a firm of licenced land surveyors. An updated bore hole location plan is appended (DWG NO. 1520A).

3.5 Petrographic and Mineragraphic Examination

Rock and mineral specimens from a wide ranging area of E.L. 81 and also the adjoining E.L. 50 have been examined by petrologist-mineralogist, P.J. Curtis (Pacminex).

This work is proving to be of considerable benefit to field staff, especially during the course of geological mapping, in being able to relate a specimen of apparent differing lithologies to type units as seen at Mt. Gunson.

It is expected that the combination of field mapping and stratigraphic drilling will do much towards gaining a greater understanding of the sedimentation on this portion of the Stuart Shelf. This information in turn is expected to have considerable bearing on the origin of the Gunson copper deposits and on theories to be used in attempts to discover further deposits.

4. EXPLORATION RESULTS

4.1 Regional

With the widespread drilling that has been carried out in the past four months, knowledge of the distribution of rock units overlying the Pandurra Formation has increased.

The Woocalla Dolomite and its deep water equivalent, a black shaley siltstone, has now been traced southward from under Mount Gunson, under the Oakden Hills, to south of Monalena, with only a narrow ridge separating the western dolomite-subcrop from the Pernatty Lagoon dolomite occurrences. It has been further identified from the Woomera No. 1 Bore as being present below Woomera township at a depth of 308 metres to 454.5 metres.

The orientation geophysical and geochemical investigations over the Cattle Grid deposit may prove to have great significance to the exploration of E.L. 81, especially over those areas that are extensively covered by wind blown sands.

4.2 Cattle Grid

Proving up and confirmation of ore reserve figures and metallurgical studies have been the main aim of the exploration activity over the Cattle Grid deposit. To this end two shafts, sunk within M.C. 184 to a depth of 50 m in one case and 45 m in the second case, have been bulk sampled for metallurgical trials by AMDEL and Pacminex/CSR metallurgical staff. In addition, mapping and sampling of the shaft walls has been carried out. Four drives were developed from the No. 1 shaft towards four adjacent diamond drill holes. These drives were used to compare drill core assay values against larger bulk samples from the shafts and drives. Although these shafts and drives were only of small size with resultant small tonnage extraction, the data of six diamond holes compared favourably with the bulk samples.

Additionally three diamond drill holes were collared from the lowermost two drive levels, extending horizontally NE, SW and NW away from No. 1 shaft. Assay results from 0.5 m sample intervals showed that in the lower portions of the ore zone values are patchy and range from waste to high grade. These drill holes were designed to test the lowermost portions of the ore zone. The drilling suggests an irregular floor to the ore zone rather than lack of continuity in the main body of the ore zone as intercepted in drill holes. It has been noted that the upper portion of the ore zone is controlled by horizontal fracture systems probably related to earlier weathering of joints and subsequent enlargement of fractures during sulphide crystallisation. The lower portion of the ore zone gradually changes to one where sub-vertical joints, often of considerable length and of varying width, are the main loci of sulphide deposition. This lower zone is that into which the horizontal diamond

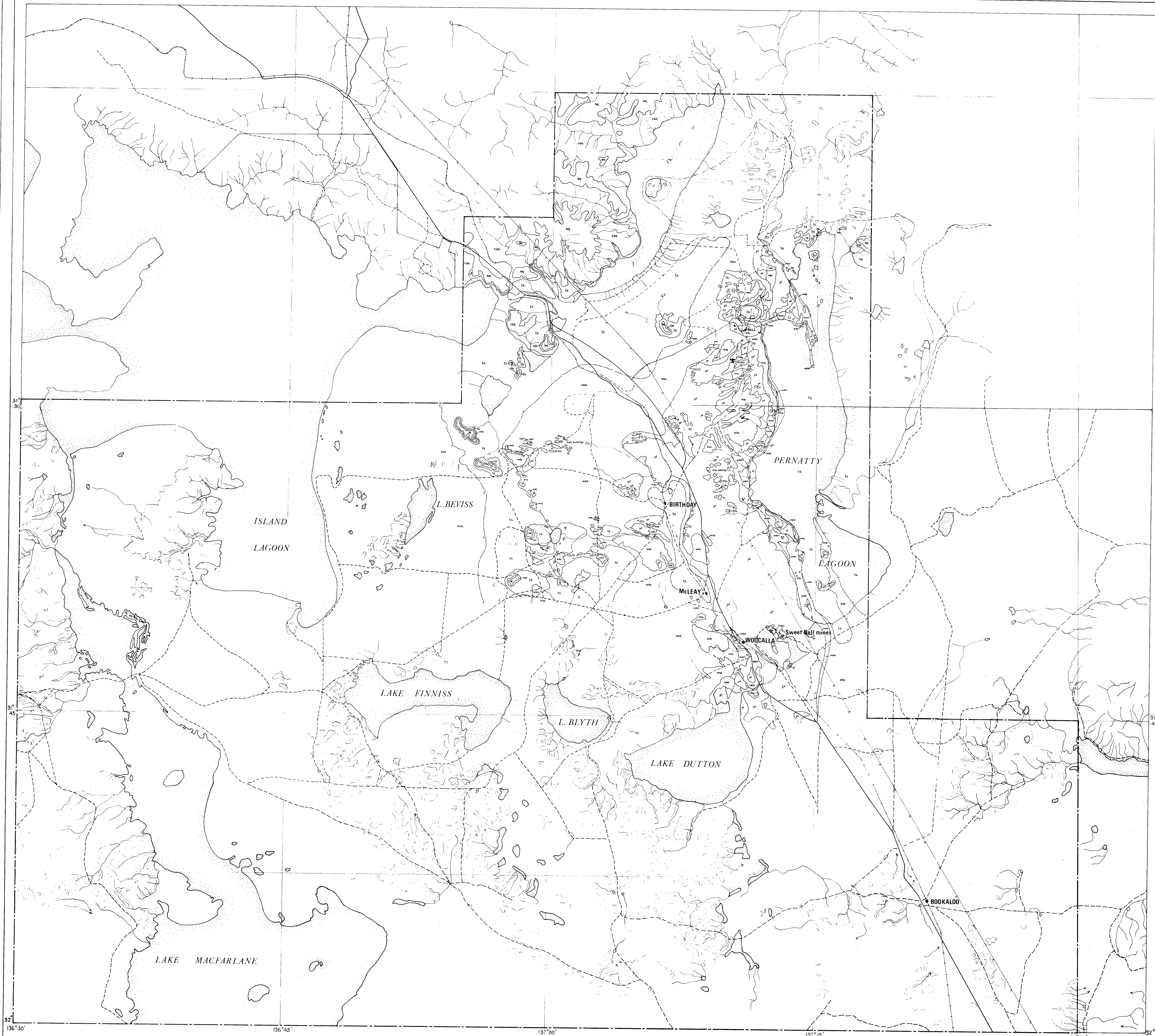
drill holes were collared. "Ore" in this zone is expected to be rather irregular in distribution and will make an irregular floor to a pit.

5. EXPENDITURE

Expenditure for the period 1st May, 1973, to 30th September, 1973, amounted to \$194,477. This total does not include costs of exploration and investigations within M.C. 184 and M.C.'s 251 to 279 inclusive.

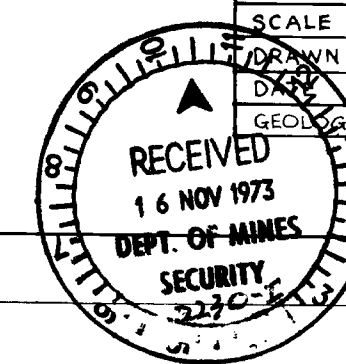
	\$
Geological	65,231
Geochemical	3,968
Geophysical	19,361
Drilling and Assaying	43,115
Technical Studies	<u>62,802</u>
Total	<u>\$ 194,477</u>

J.M. HEAPE
(EMP)

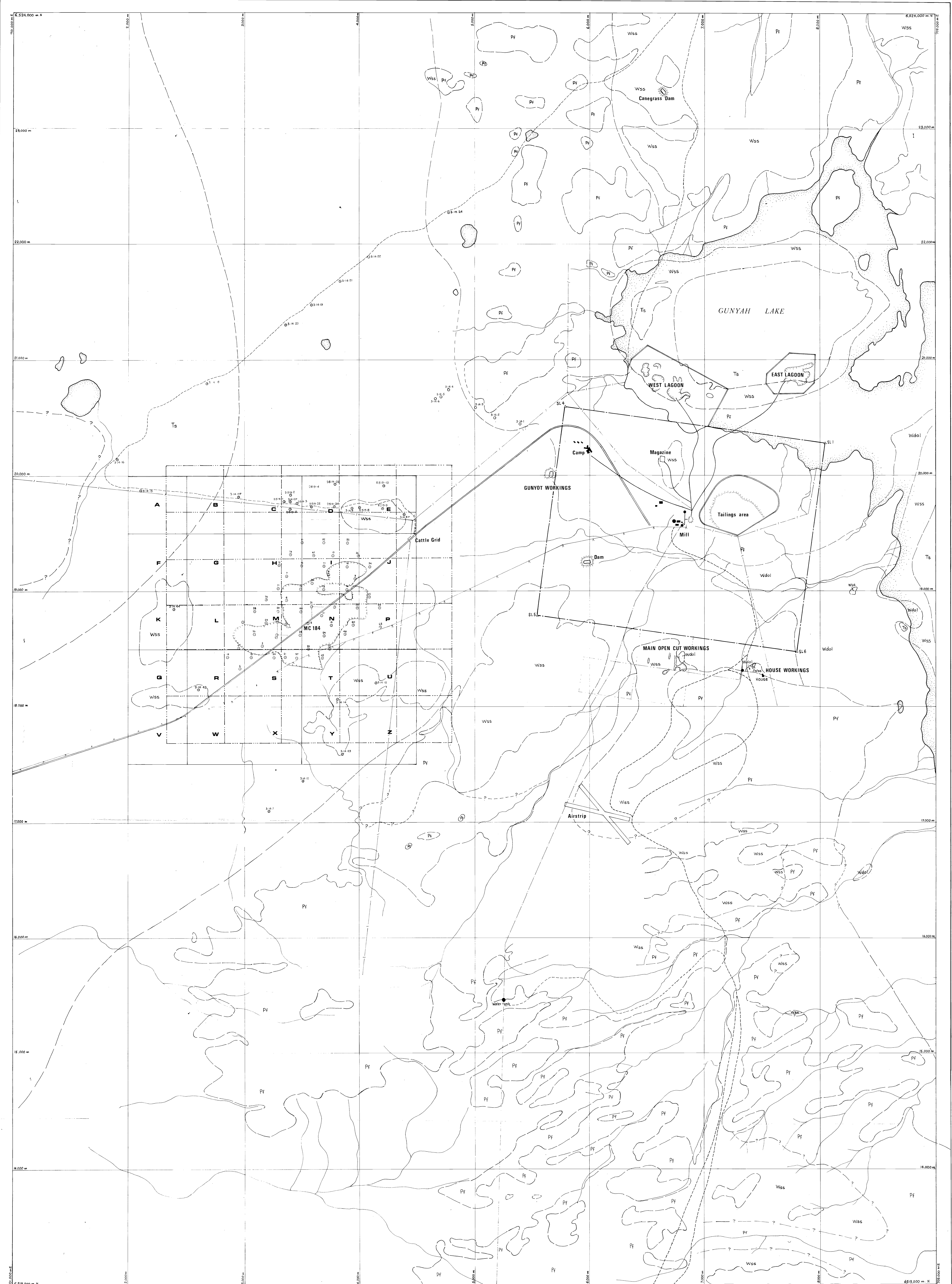


BASED ON 1:250,000 TOPOGRAPHICAL SHEETS "TORRENS" AND "PORT AUGUSTA"

PACMINEX PTY. LIMITED	
MOUNT GUNSON MINES PTY. LTD.	
E.L. 81 MT. GUNSON AREA BEDROCK LITHOLOGY	
PROJECT 401	
SCALE 1:125,000	DRAWING NO.
C.J.L. / P.K.	
FEB. 1973 / OCT. 1973	
PC / ST / J.H. / K.M.	
16 NOV 1973	1565
DEPT. OF MINES	
SECURITY	



2330(I)-1



Ts TREGOLANA SHALE
Wdol WOODCULLA DOLOMITE
Wss WHYALLA SANDSTONE
Pf PANDURRA QUARTZITE

REFERENCE

ROADS AND TRACKS
POWER LINE
WATER PIPELINE
FENCES
M.P.L. BOUNDARY
M.L. BOUNDARY

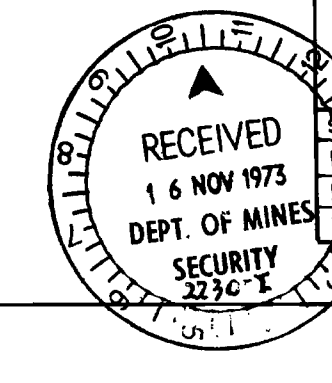
PACMINEX PTY. LIMITED

MOUNT GUNSON MINES PTY. LTD.

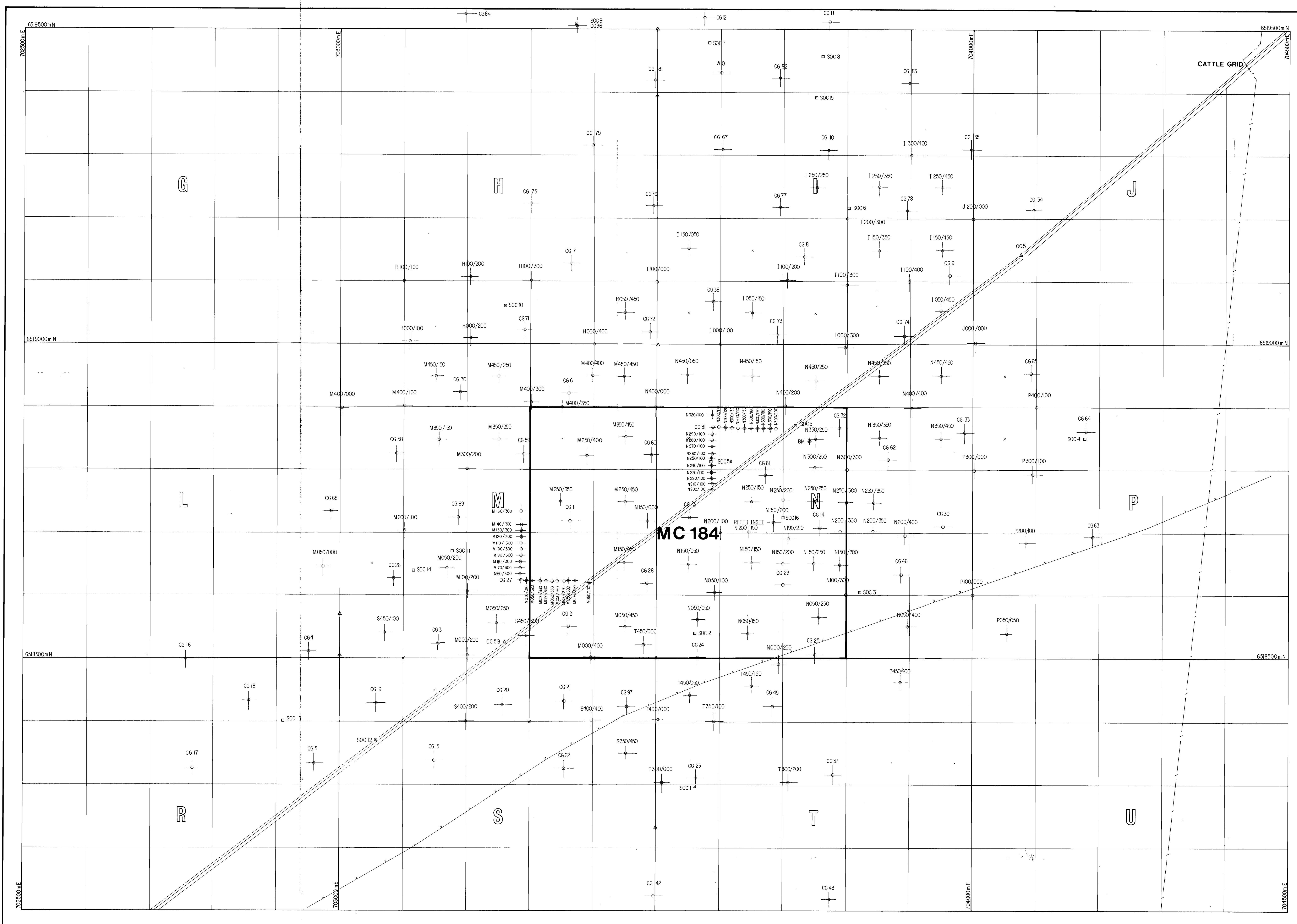
OUTCROP AND BEDROCK GEOLOGY

SHEET 1

1:10,000
C.J.L.
16 NOV 1973
MAY 18, 1973
NOV 1973
1521a



2330(I)-2



REFERENCE

- Traverse
- == Road
- v- Power line
- SOC Survey base
- Δ Survey datum peg
- ⊕ Drill hole
- × Proposed drill hole
- /- Fence
- ⊕ Drill hole, position approx.

INSET

PACMINEX PTY. LIMITED

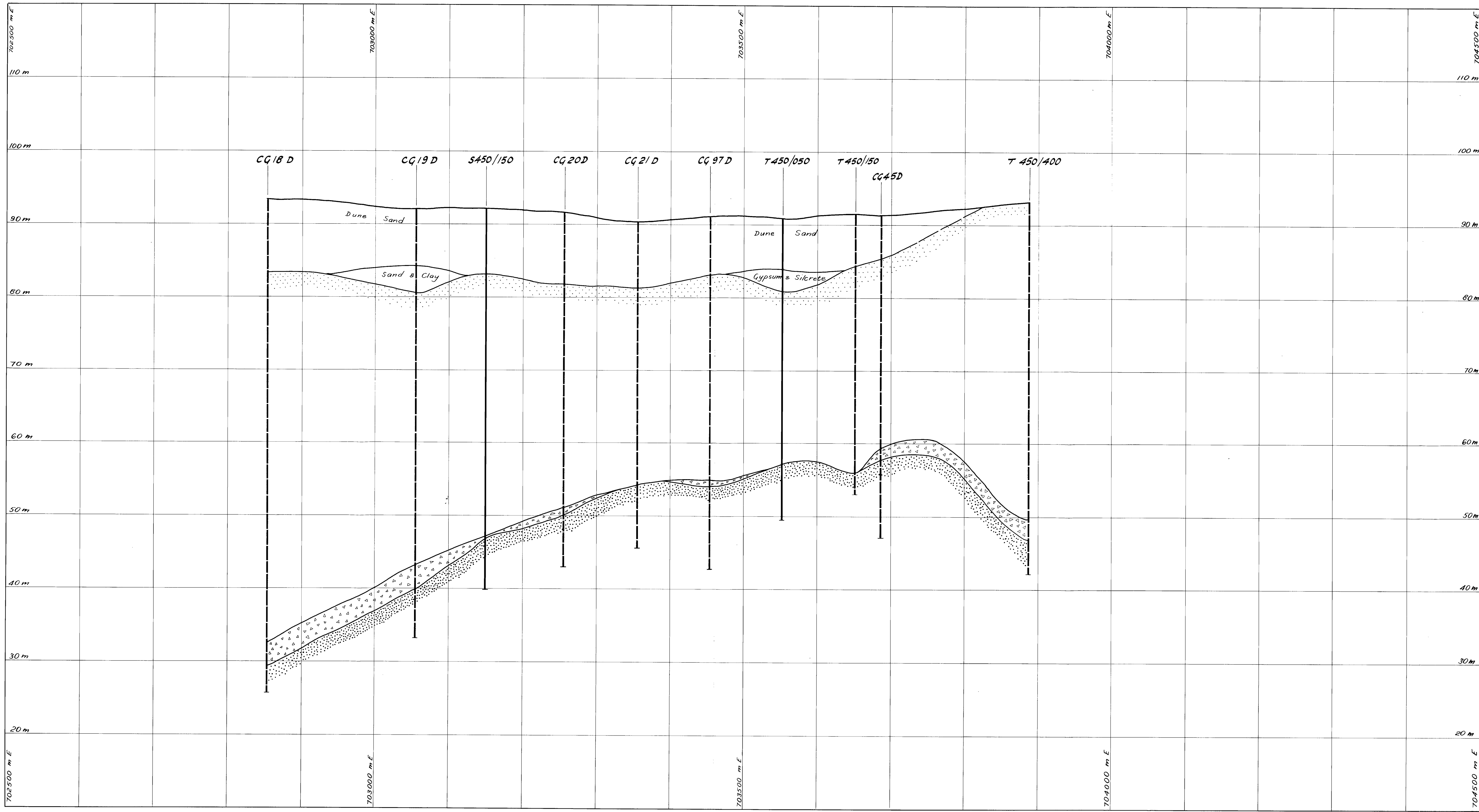
MOUNT GUNSON MINES PTY. LTD.

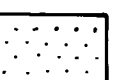
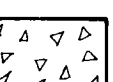
DRILL HOLE LOCATION PLAN

CATTLE GRID AREA
Sheet 5

SCALE	1: 2500	DWG. No.	1520A
DRAWN	P.R. /		
DATE	May 1973		
REVISED	Sept. 1973		

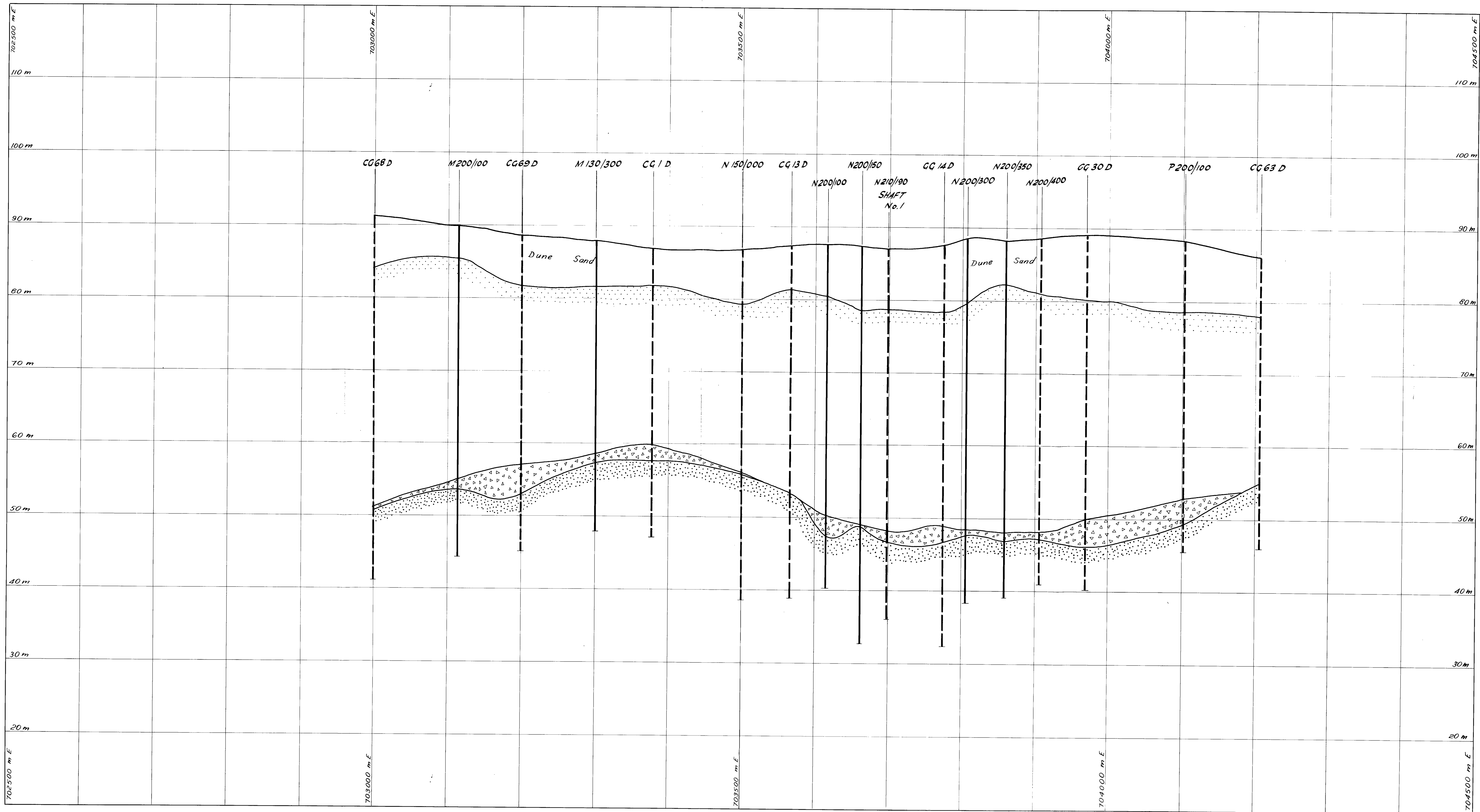
2330(I)-3

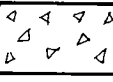


-  Whyalla sandstone
-  Transition zone
-  Pandurra formation
-  Drill hole on section
-  Drill hole projected onto section



PACMINEX PTY. LIMITED	
MOUNT GUNSON MINES PTY. LTD.	
CATTLE GRID AREA	
CROSS SECTION 65/8450mN	
Scale Horiz. 1:2500 Vert. 1:250	1526 a
Drawn P.S.I. / P.Y.	
Date Nov. 1973	
Revised	

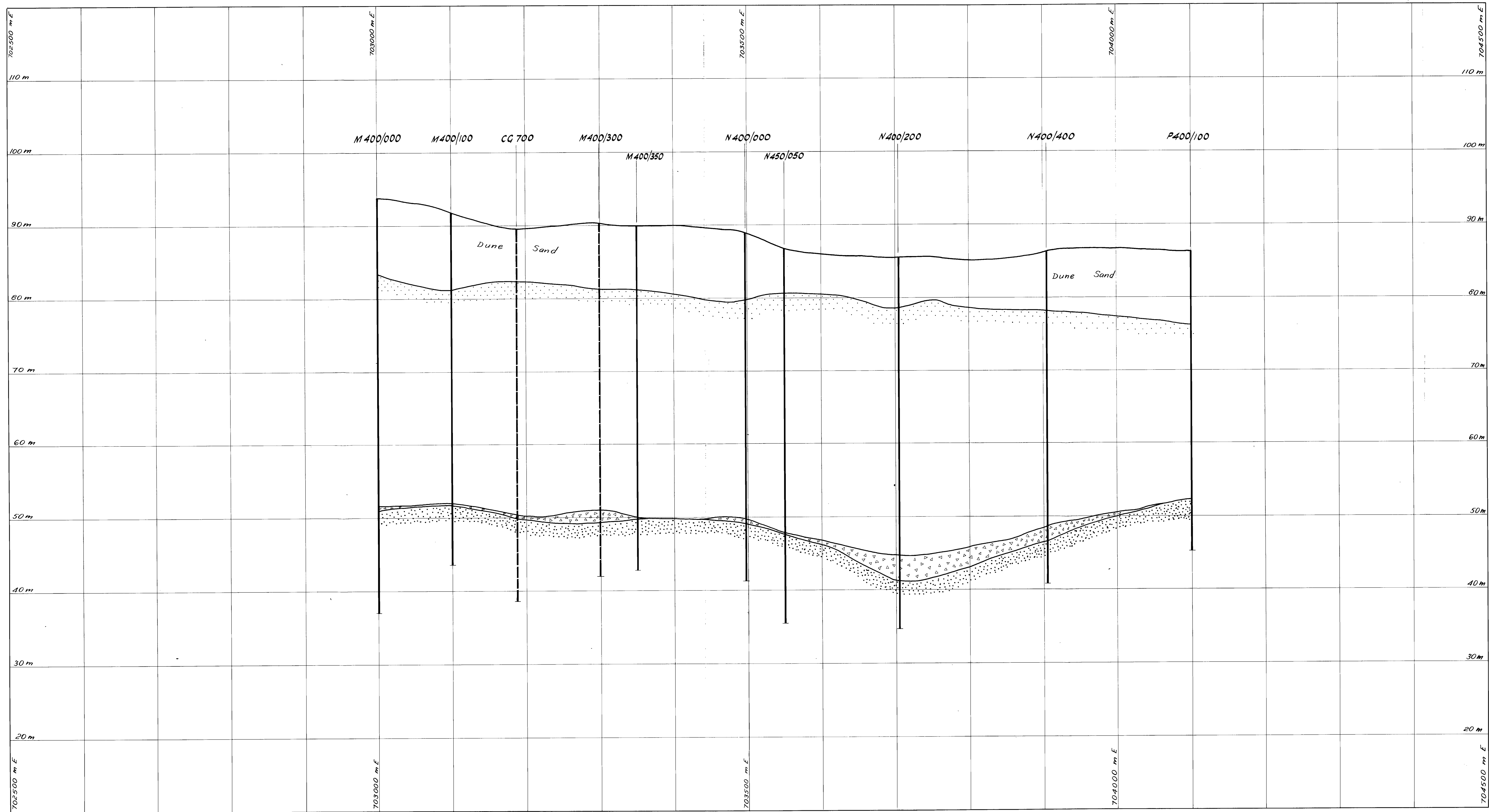


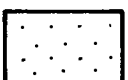
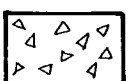
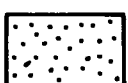
-  Whyalla sandstone
-  Transition zone
-  Pandurra formation
-  Drill hole on section
-  Drill hole projected onto section



PACMINEX PTY. LIMITED	
MOUNT GUNSON MINES PTY. LTD.	
CATTLE GRID AREA	
CROSS SECTION 65/8700 m N	
Scale Horiz 1:2500 Vert 1:250	1526 b
Drawn P.S.I./P.Y.	
Date Nov 1973	
Revised	

2330 (I)-5

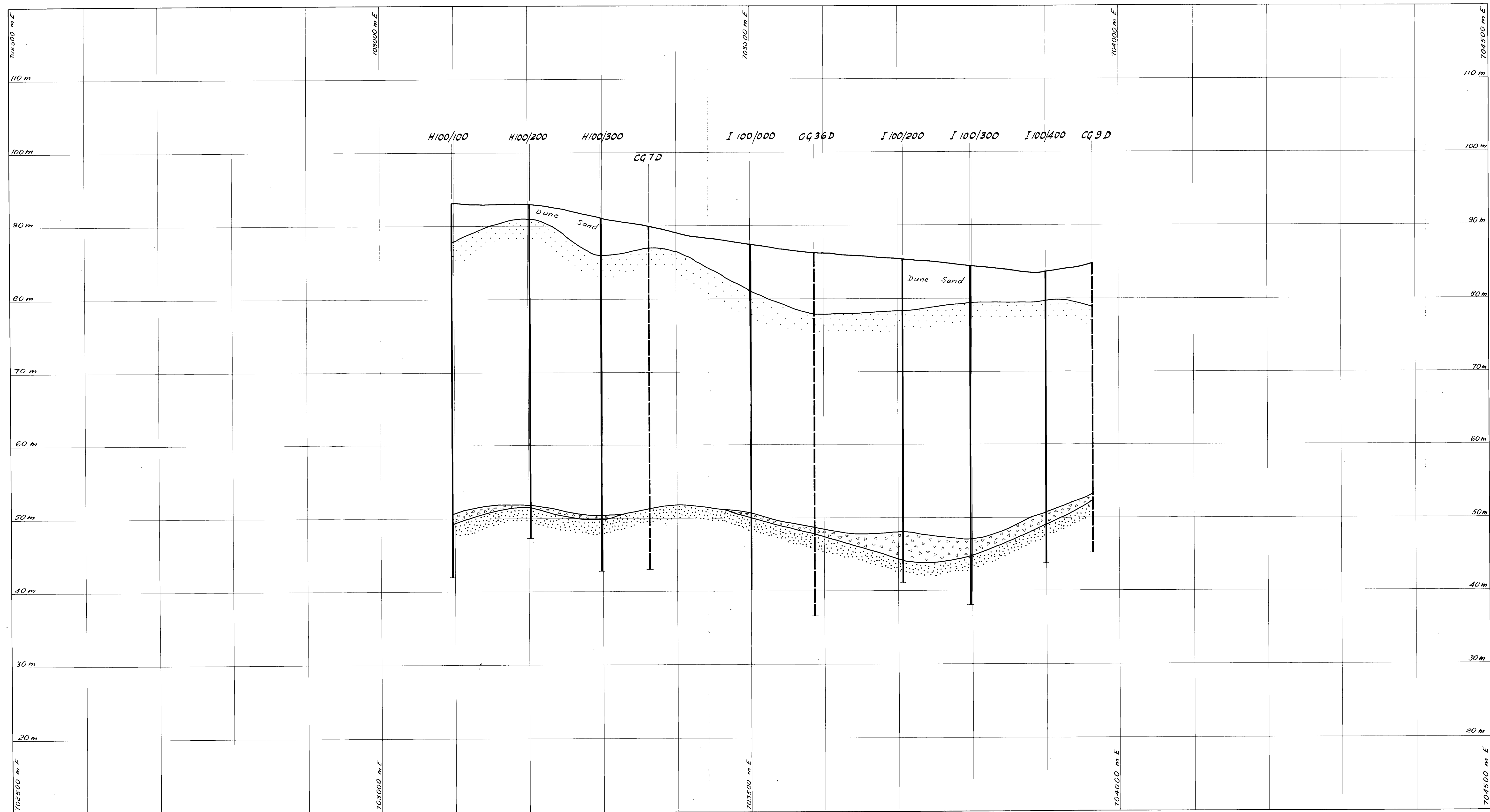


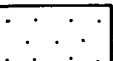
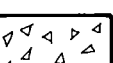
-  Whyalla sandstone
-  Transition zone
-  Pandurra formation
-  Drill hole on section
-  Drill hole projected onto section



PACMINEX PTY. LIMITED	
MOUNT GUNSON MINES PTY. LTD.	
CATTLE GRID AREA	
CROSS SECTION 65/8900 m N	
Scale Horiz. 1:2500 Vert. 1:250	1526 c
Drawn P.S.I. / P.Y.	
Date Nov 1973	
Revised	

2330(I)-6



-  Whyalla sandstone
-  Transition zone
-  Pandurra formation
-  Drill hole on section
-  Drill hole projected onto section



PACMINEX PTY. LIMITED	
MOUNT GUNSON MINES PTY. LTD.	
CATTLE GRID AREA	
CROSS SECTION 65/9100 m N	
Scale Horiz 1:2500 Vert 1:250	1526d
Drawn PSI / PY	
Date Nov 1973	
Revised	

2330(I)-7

Ref. No. PMR. 97/73

PACMINEX PTY LIMITED

RESULTS OF EXPLORATION ACTIVITIES
IN THE OAKDEN HILLS-LAKE MACFARLANE AREA

MT. GUNSON
September, 1973.

K.J. MAIDEN

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0018

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PLANS

PMX DWG NO.

47	REGIONAL GEOLOGY (1:125,000)	
1562	AEROMAGNETIC INTERPRETATION (1:125,000)	
1551	BIRTHDAY-MT. MOSELEY : GEOLOGY (1:30,000)	
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1. SUMMARY

Mapping and drilling in the Birthday-Mt. Moseley-Oakden Hills area has revealed that the main ridge of Pandurra Formation (stretching along the western side of Pernatty Lagoon) extends into the Oakden Hills area as a complex of buried ridges separated by shallow valleys. To the north, west and south of the ridge complex, the post-Pandurra surface slopes away into broad basins of deposition, infilled with the black shale facies of the Woocalla Dolomite.

A scout drilling programme between Oakden Hills and Lake MacFarlane has shown that the black shale basin extends westward to Lake MacFarlane. On the western side of the lake, the post-Pandurra surface rises to another Pandurra ridge.

Target areas for further exploration are the slopes of the Pandurra ridges where the Woocalla Dolomite forms an on-lap structure onto the Pandurra Formation. In particular, on the basis of high copper values, three areas have been delineated for follow-up work.

2. PHYSIOGRAPHY

The area consists mainly of a sandy plain with generally sparse vegetation and scattered low rock outcrops. Across the northern part of the area, stretching from the Transcontinental Railway Line to Island Lagoon, the sandy plain grades into sand dune country with quite a thick covering of stunted trees and almost no outcrops. Salt lakes occupy a large portion of the area. In the east, the land rises slightly to the ridge of Pandurra Quartzite which extends along the western shore of Pernatty Lagoon. Similarly, in the west the land rises to another ridge of Pandurra Formation on the western shore of Lake MacFarlane. Prominent outcrops of Whyalla Sandstone occur along the western edge of Island Lagoon and Lake Finniss. In the centre of the area, four mesas, capped by Simmens Quartzite, form prominent features.

Station tracks provide good access to most parts of the area. Cross country travel in four-wheel drive vehicles is no problem in the sandy plain and on the Pandurra ridges, but becomes almost impossible in the sand dune country.

3. GENERAL STRATIGRAPHY

The Adelaidean sediments which outcrop in the Mt. Gunson-Pernatty Lagoon area can all be identified in the Oakden Hills-Lake MacFarlane area. Table 1 gives the maximum thicknesses of the units as intersected in drill holes and as estimated in the field.

Table 1

Estimated Maximum Thicknesses of Units
in the Oakden Hills-Lake MacFarlane Area

Unit	Maximum Thickness
Simmens Quartzite	40 m
Corraberra Sandstone	40 m
Tregolana Shale	140 m
Whyalla Sandstone	100 m
Woocalla Dolomite	130 m
Pandurra Formation	>100 m

4. PREVIOUS WORK

The geology of the area has been mapped on a regional scale by geologists of the Department of Mines, and published in the 1964 edition of the Torrens 1:250,000 map. The regional geology was described by R.K. Johns in a 1968 publication entitled "Geology and Mineralisation of the Andamooka-Torrens Area".

Aeromagnetic maps are available on scales of 1:63,360 and 1:250,000. An interpretation of the aeromagnetic results has been carried out by Austral Exploration Services (see Aeromagnetic Interpretation Map).

Noranda Australia Ltd held leases over portion of the area and carried out regional mapping (on a 1:125,000 scale) and geochemical sampling. This was followed by rotary and diamond drilling. Three of the Noranda drill holes (LD 1, LD 2 and LD 3) are shown on the 1:125,000 Regional Geology Map.

Table 2

0024

Drill Hole Summary

Hole Number	Date Drilled	Approx. Co-ordinates		Hole Depth	Depth to Stratigraphic Units				
		Northing	Easting		Tregolana	Whyalla	Woocalla	Pandurra	
<u>MacFarlane Area :</u>									
MF 1	19/6/73	489,500	650,000	94.5	-	14	29.5	38	
MF 2	21/6/73	488,500	655,000	94.5	-	25	29	71	
MF 3	21/6/73	489,500	653,000	15	-		Not Reached		
MF 4	22/6/73	496,000	657,000	91.5	-	4	20	89	
MF 5	23/6/73	488,000	665,000	96	-	7	16	95	
MF 6	24/6/73	490,000	675,000	105	-	5	99	Not Reached	
<u>Oakden Hills Area :</u>									
OK 1	12/5/73	505,000	700,500	72	4	49	Not Present	70	
OK 2	13/5/73	500,500	699,000	25.5	-	0	Not Present	24	
OK 3	13/5/73	500,000	693,500	47.5	4	5.5	Not Present	43	
OK 4	13/5/73	507,000	694,000	41	-	8	Not Present	39	
OK 5	14/5/73	510,000	693,000	22	-	5	Not Present	21	
OK 6	14/5/73	510,000	690,000	9	-	1	Not Reached		
OK 7	15/5/73	507,500	686,000	85	3	35	Not Reached		
OK 8	15/5/73	499,000	690,000	81	-	0	53	74	
OK 9	26/6/73	496,000	685,000	108	3	34	Not Reached		
OK 10	27/6/73	496,500	703,000	42	2	19.5	26	38.5	
OK 11	29/6/73	512,000	698,500	45	-	15	32	41.5	
OK 12	29/6/73	511,000	699,000	31.5	-	9	Not Present	19.5	